



Thermal Analysis Course of Solar PV Module

THE ANALYSIS METHODOLOGY

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The Thermal Analysis of PV Module:

Energy, from solar radiation incident on the PV panel, is generally leaves the system as *generated power*, *optical loss* and *thermal losses*. Heat transfer processes including convection, and radiation from the top and bottom layers to the surrounding environment all contribute to the overall thermal loss. Reflective radiation by the glass (or ETFE) layer represent the major optical loss. In addition, the generated power is also included in the energy balance of PV module as will be explained in this lecture

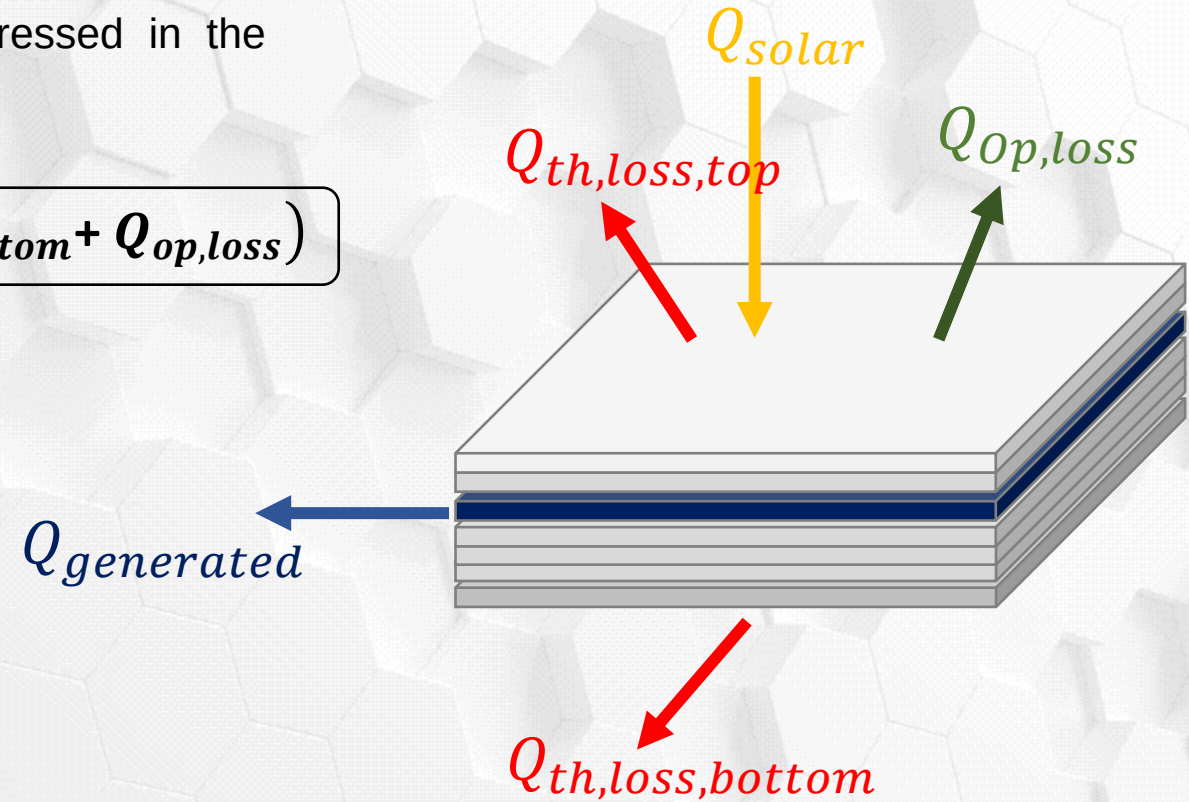
The Thermal Analysis Assumptions:

- All material properties used are assumed to be isotropic and independent of temperature.
- The reflectivity of the various materials is considered significant to the analysis. Solar irradiation that is not reflected off any surface is fully transmitted to the layer below.
- The transmissivity of the EVA is taken to be unity.
- The internal reflection between the upper EVA and the solar cells is considered and other internal reflections between layers are negligible.
- The part of the solar irradiation that is neither reflected by the cell nor converted to electricity is converted to thermal energy.
- The ambient temperature is equal on every area exposed to the environment.

The Energy Balance:

The energy balance of the solar PV panel is expressed in the following form;

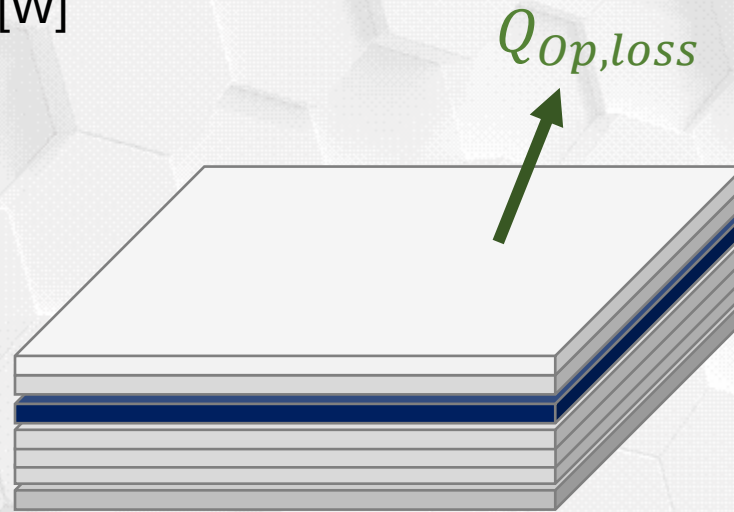
$$Q_{generated} = Q_{solar} - (Q_{th,loss,top} + Q_{th,loss,bottom} + Q_{op,loss})$$



The Optical loss ($Q_{op,loss}$):

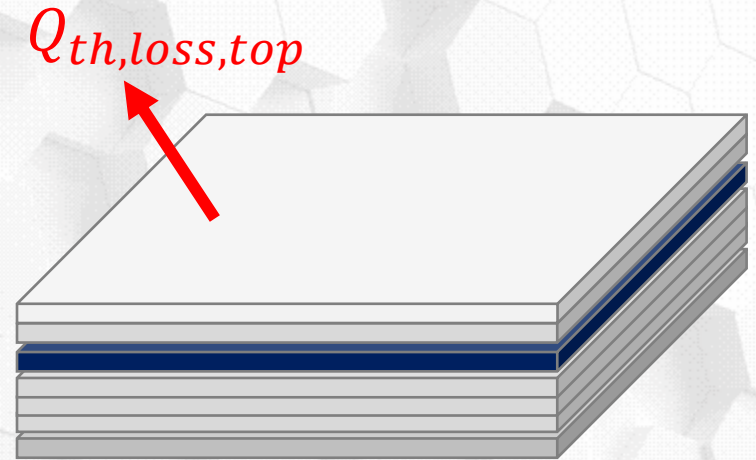
The ETFE layer reflectivity (r_{ETFE}) is the primary contributor to optical heat loss because its impact results in making a part of incident solar radiation to be reflected back to the environment. This type of loss can be calculated using the relation shown below:

$$Q_{op,loss} = r_{ETFE} Q_{solar} \quad \dots [W]$$



The top surface thermal losses ($Q_{th,loss,top}$):

Convection and radiation losses are the two types of thermal losses that exist on the top surface of the PV panel. These losses flow out from the surface of the glass (ETFE) that receives the focused flux. The convection losses are a combination of free (natural) and forced convection. There is also thermal radiation loss between the glass layer and the environment.



$$Q_{th,loss,top} = Q_{Forced,conv,top} + Q_{natural,conv,top} + Q_{rad,top}$$

The top surface thermal losses ($Q_{th,loss,top}$):

$$Q_{th,loss,top} = Q_{Forced,conv,top} + Q_{natural,conv,top} + Q_{rad,top}$$

The forced convective coefficient can be estimated from the correlation

$$Nu_{F,t} = \frac{h_{F,t}L}{k} = 0.13Re_L^{0.703}(1 + \sin \beta)^{0.38}$$

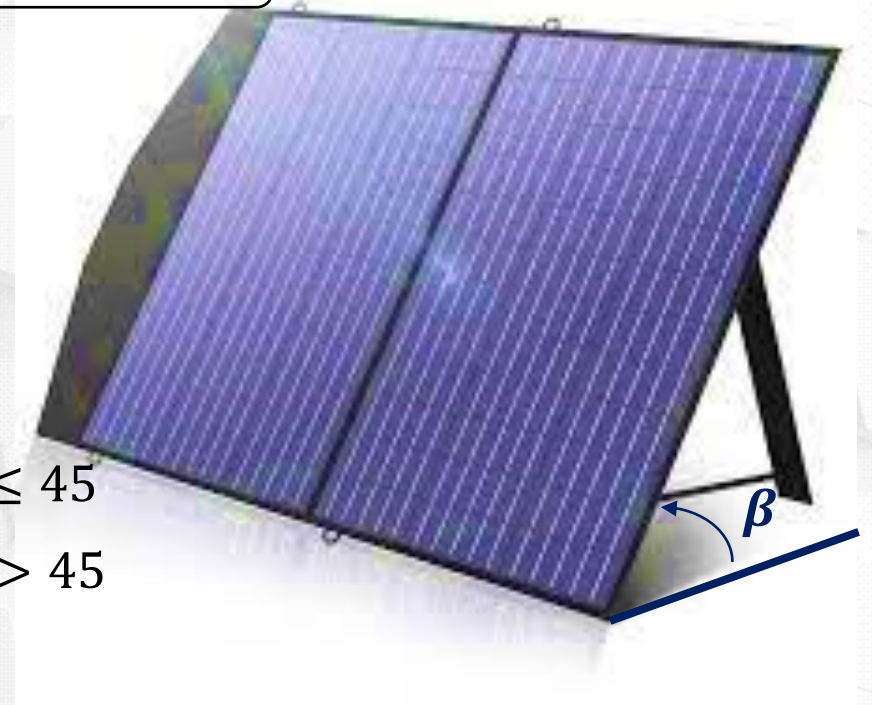
And the natural convective coefficient can be obtained from the correlation

$$Nu_{N,t} = \frac{h_{N,t}L}{k} = 0.7386Ra_L^{0.1826}(1 + \cos \beta)^{-0.4575} \quad \dots \beta \leq 45$$

$$= 5.4412 Ra_L^{0.1102}(1 + \cos \beta)^{-0.085} \quad \dots \beta > 45$$

Therefore, the mixed convective coefficient is followed the formula:

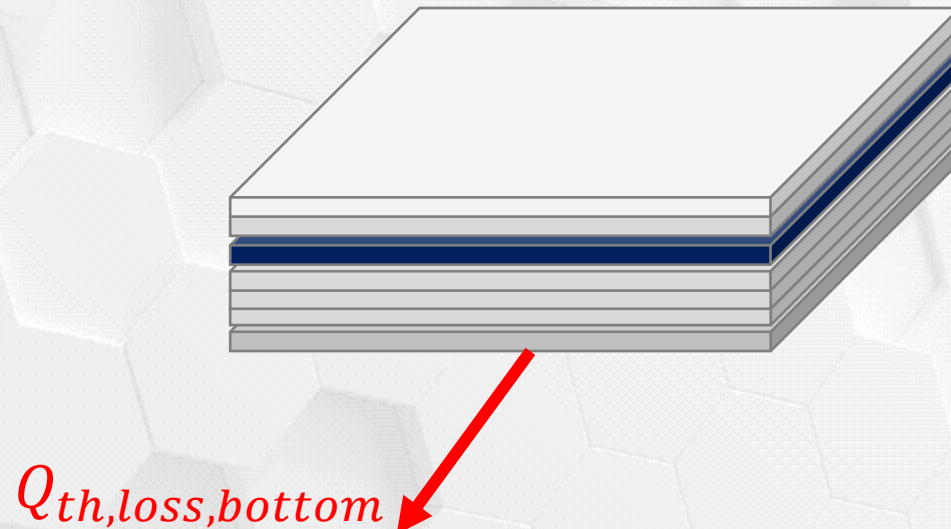
$$Nu_{M,t}^3 = Nu_{F,t}^3 + Nu_{N,t}^3$$



The bottom surface thermal losses ($Q_{th,loss,bottom}$):

The thermal losses from the bottom surface can be expressed using the same relation that's been given for the top surface.

$$Q_{th,loss,bottom} = Q_{Forced,conv,bottom} + Q_{natural,conv,bottom} + Q_{rad,bottom}$$



The bottom surface thermal losses ($Q_{th,loss,bottom}$):

$$Q_{th,loss,bottom} = Q_{Forced,conv,bottom} + Q_{natural,conv,bottom} + Q_{rad,bottom}$$

The forced convective coefficient can be estimated from the correlation

$$h_{F,b} = h_{F,t} \left(\frac{90^\circ - \beta}{90^\circ} \right)$$

And the natural convective coefficient can be obtained from the correlation

$$\begin{aligned}
 Nu_{N,b} = \frac{h_{N,b}L}{k} &= 0.27Ra_L^{0.25} & \dots \beta = 0 \\
 &= 0.7386Ra_L^{0.1826}(1 + \cos \beta)^{-0.4575} & \dots 0 < \beta \leq 45 \\
 &= 5.4412 Ra_L^{0.1102}(1 + \cos \beta)^{-0.085} & \dots \beta > 45
 \end{aligned}$$

Therefore, the mixed convective coefficient is followed the formula:

$$Nu_{M,b}^3 = Nu_{F,b}^3 + Nu_{N,b}^3$$

References:

- A. R. EL-SHAMY, R. Y. SAKR*, N. S. BERBISH, AND M. H. MESSRA, "EXPERIMENTAL AND NUMERICAL STUDY OF FORCED CONVECTION HEAT TRANSFER FROM AN INCLINED HEATED PLATE PLACED BENEATH A POROUS MEDIUM", Al-Azhar University Engineering Journal, JAUES, Vol. 2, No. 4, Apr. 2007.
- Shuang-Ying Wu, Ying-Ying Wu, Lan Xiao, and Zhen Yang, "Experimental study of natural convection heat transfer from a nonuniformly heated flat plate simulating PV panel", Journal of Mechanical Science and Technology, Vol 32, no.1, pp. 423-432, 2018.
- F. P. INCROPERA, D. P. DEWITT, T. L. BERGMAN, and A. S. LAVINE, "Fundamentals of Heat and Mass Transfer", 6th Edition, 2015.